Intern AWS Agenda

IT intern onboarding itinerary tailored specifically for an IT department with a focus on **Azure and AWS cloud services**. It's designed as a **1-week program**, but you can stretch or compress it depending on your schedule.

⛅ IT Intern Cloud Onboarding Itinerary (1–Week Program)

Below is a well-organized teaching plan with topics, objectives, and suggested activities.

📅 Day 1 — Welcome & Foundations

1. Introductions & Orientation

- Meet the IT team and leadership
- Overview of departmental structure (Helpdesk, Networking, Security, Cloud, etc.)
- Review organizational policies:
  - Security policies
  - Acceptable use
  - Change management
  - Ticketing procedures

2. IT & Cloud Fundamentals

- What is Cloud Computing? (IaaS, PaaS, SaaS)
- Overview of on-prem vs cloud
  - Basic components:
    - Compute
    - Networking
    - Storage

📅 Day 2 — AWS Fundamentals

1. AWS Overview

- AWS Management Console walkthrough
- Regions & Availability Zones
- IAM basics
- Resources
- Cost awareness & tagging strategies

2. Core AWS Services

- **Compute**: EC2, Lambda
- **Networking**: VPC, Security Groups
- **Storage**: S3, EBS
- **Identity**: IAM Users, Roles, Policies

3. Hands-On Lab

- Launch an EC2 instance
- Configure security groups
- S3 bucket creation & file uploads
- Attach IAM roles to EC2

📅 Day 4 — DevOps, Automation & Cloud Tools

1. Infrastructure as Code (IaC)

- Terraform basics
- AWS CloudFormation overview
  - JSON
  - YAML

2. DevOps Concepts

- CI/CD pipelines (AWS CodePipeline or GitHub Actions)
- Source control workflows (Git: branching, PRs, merges)

3. Monitoring & Logging

- AWS CloudWatch
- Alerts and dashboards

4. Hands-On Lab

- Create a small Terraform/CloudFormation project to deploy a EC2 or S3 bucket

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**📅 Day 5 — Security, Best Practices

1. Cloud Security Basics

- Zero Trust
- Identity protection (MFA, Conditional Access)
- Key Vault / AWS KMS
- Network segmentation
- Least privilege and RBAC/IAM best practices

2. Cost Optimization

- AWS Trusted Advisor
- Rightsizing resources
- Stopping/Terminating unused EC2

3. Capstone Project

Interns pick one:

- Build a small multi-tier lab in AWS
- Deploy a secure web app on AWS
- Implement a monitoring dashboard
- Create a Terraform-based deployment

4. Presentation

Interns present:

- What they built
- What they learned
- What challenges they encountered

🎓 Bonus Topics (Optional for Extended Training)

Cybersecurity

- SOC operations overview
- Incident response workflow
- Defender for Cloud / AWS GuardDuty

Networking

- DNS fundamentals
- VPNs, ExpressRoute, Direct Connect

☁️ IT Intern Cloud Training

AWS Hands-On Labs, Quizzes & Assessments

🧪 AWS Hands-On Labs (Progressive Difficulty)

AWS Lab 1 — AWS Account & IAM Fundamentals

Objective: Teach secure access and identity management.

Topics Covered

- AWS Console navigation
- IAM users, groups, roles
- MFA
- Least privilege principle

Lab Steps

1. Log into AWS Console
2. Create an IAM user
3. Assign to an IAM group
4. Attach a managed policy (ReadOnlyAccess)
5. Enable MFA
6. Test permissions by accessing services

✅ Success Criteria

- User cannot create resources
- MFA is enforced
- Intern explains why root account should not be used

AWS Lab 2 — EC2 Compute & Networking

Objective: Deploy and secure a virtual server.

Topics Covered

- EC2 instances
- AMIs
- Security Groups
- SSH/RDP access

Lab Steps

1. Launch EC2 instance (Amazon Linux or Windows)
2. Select instance type
3. Configure Security Group:
   - Allow SSH (22) or RDP (3389)
4. Generate key pair
5. Connect to the instance

✅ Success Criteria

- Instance is reachable
- Security group rules are minimal
- Intern can explain public vs private IPs

AWS Lab 3 — VPC & Networking Basics

Objective: Understand AWS networking architecture.

Topics Covered

- VPC
- Subnets (public/private)
- Internet Gateway

- Route tables

Lab Steps

1. Create a custom VPC
2. Create public and private subnets
3. Attach Internet Gateway
4. Update route tables
5. Deploy EC2 into public subnet

✅ **Success Criteria**

- EC2 in public subnet is reachable
- EC2 in private subnet is not

AWS Lab 4 — S3 Storage & Permissions

Objective: Learn object storage and access control.

Topics Covered

- S3 buckets
- Bucket policies
- Versioning
- Encryption

Lab Steps

1. Create an S3 bucket
2. Enable versioning
3. Upload files
4. Enable default encryption
5. Configure bucket policy (read-only access for IAM role)

✅ **Success Criteria**

- Files are versioned
- Bucket access is restricted
- Intern understands public vs private buckets

AWS Lab 5 — IAM Roles & EC2 Access

Objective: Teach role-based access instead of credentials.

Topics Covered

- IAM roles
- Instance profiles
- Secure service access

Lab Steps

1. Create IAM role with S3 read access
2. Attach role to EC2 instance
3. Access S3 from EC2 without credentials

✅ Success Criteria

- No access keys used
- Access works via role

AWS Lab 6 — Monitoring with CloudWatch

Objective: Understand monitoring and alerting.

Topics Covered

- Metrics
- Logs
- Alarms

Lab Steps

1. View EC2 metrics
2. Create CPU alarm
3. Trigger alarm using stress test
4. Review logs

✅ Success Criteria

- Alarm triggers
- Intern explains monitoring importance

AWS Lab 7 — Automation with Terraform (Optional Advanced)

Objective: Introduce Infrastructure as Code.

Topics Covered

- Terraform basics
- Providers
- State

Lab Steps

1. Initialize Terraform
2. Deploy EC2 or S3
3. Modify resource
4. Destroy resources

✅ Success Criteria

- Resources created via code
- Intern understands idempotency

📝 Quizzes (Knowledge Checks)

Daily Quiz Format

- 5–10 questions
- Multiple choice + short answer
- Pass score: **80%**

Sample AWS Quiz Questions

1. What is the purpose of an IAM role?
A. Store passwords
B. Provide temporary permissions ✅
C. Replace security groups
D. Manage billing

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2. What service provides object storage in AWS?
A. EBS
B. EFS
C. S3 ✅
D. Glacier

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3. Which AWS construct controls inbound and outbound traffic?
A. Route Tables
B. Security Groups ✅
C. IAM Policies
D. NACL only

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4. True or False:
You should always use the root account for daily tasks.
❌ **False**

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5. What happens if you delete an EC2 instance?

- Instance is terminated
- Root volume may be deleted
- Data loss occurs if not backed up

✅ **Expected explanation**

📊 Assessments & Evaluation

Mid-Week Assessment (Day 3)

Format

- 20-question quiz
- 1 practical lab

Lab Scenario

> Deploy an EC2 instance in a secure VPC and upload a file to S3 using IAM roles.

Final Capstone Assessment (Day 5)

Intern must:

- Design a basic AWS solution
- Deploy at least:
  - EC2
  - S3
  - IAM roles
  - CloudWatch alarm

Deliverables

- Architecture diagram
- Security explanation
- Cost considerations
- Demo walkthrough

✅ Evaluation Criteria

Area	Weight
Security	30%
Functionality	30%
Documentation	20%
Explanation	20%